Equal-Budget Prompt Pooling Does Not Beat Self-Consistency—And Seed-Matched Controls Bound Apparent Selector Gains

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Abstract

Self-consistency (SC) is capped by a modal ceiling π_{mode}. Semantics-preserving elicitation can change plug-in modes, yet equal-budget configuration-diversified voting (CDV) need not help. On a crossed sampled-CoT grid (Qwen2.5 {1.5B,3B,7B}, Llama-3.2-3B; MATH-500, GSM8K, GSM-Symbolic; $\{c_0, c_1, c_2\}$) we find: (C1) plug-in π_{mode} spreads >3 pp on several cells (largest on GSM-Symbolic); (C2) token-matched equal-budget CDV shows no positive lift vs. SC (mean $\Delta \approx -0.7$ pp; pooled CI crosses zero); (C3) a pre-specified conditional-diversify rule gains $\approx +2.3$ pp vs. SC@N over 14 cells (CI $[+1.1, +3.7]$), but on the 6 multi-seed cells seed-substitution recovers $\approx +2.0$ pp and config−seed is ≈ -0.4 pp (CI $[-1.2, +0.3]$), so we cannot distinguish configs from seeds. An independent margin-split still beats SC; probe-then-commit costs $\approx 1.6\times$ SC@N tokens ($\approx 0.53\times$ always-3). Seed-null oracles (≈ 4.5 pp) match cross-config plug-in oracles. We do not claim elicitation recovery; the contribution is a careful CDV negative plus a protocol warning against confounding selectors with denoising.

1. Introduction

Progress in language-model reasoning is increasingly mediated by how a model selects an answer under sampling. Compute-optimal test-time scaling bins items by estimated difficulty (Snell et al., 2025); adaptive self-consistency stops when a vote looks decided (Aggarwal et al., 2023; Li et al., 2024); and analyses that explain why extra samples stop helping bound selection by the modal-hit rate π_{mode}—the fraction of items whose plurality answer is correct (Brown et al., 2024; Feng et al., 2025; Cordero-Encinar & Duncan, 2025; Bay & Yearick, 2026). More i.i.d. samples estimate the mode of a fixed answer distribution more precisely; they cannot change which class is modal (Proposition 1). That fact invites three linked questions the modal-ceiling literature leaves open for sampled chain-of-thought:

C1. Does semantics-preserving elicitation change the mode, so π_{mode} is a model–prompt property rather than a model constant?

C2. Does spreading a fixed completion-token budget across configurations (equal-budget CDV) raise π_{mode} above single-configuration SC?

C3. Over a multi-config pool, does a pre-specified held-out rule beat SC@N—and does that lift survive seed-substitution and SC@3N controls that match sample count?

Prompt sensitivity of item difficulty is already established on log-prob-scored suites (Romanou et al., 2026; Messing, 2026) and for unstable hardness labels (Zhou et al., 2026b). Those works do not measure whether sensitivity reaches the mode under sampled CoT, nor do they require seed-matched controls before attributing multi-prompt gains to elicitation.

What we find. C1: plug-in spreads can exceed a c_0 seed-null; only 2/6 multi-seed cells clear both >3 pp and a seed-permutation test at α=0.05 (largest Symbolic cells still await seeds). C2: equal-budget CDV shows no positive lift ($\Delta \approx -0.7$ pp; CI crosses zero). C3: conditional diversify gains $\approx +2.3$ pp vs. SC@N, but config−seed and config−SC@3N CIs both cross zero on controlled cells; margin-split and drop-c_2 ablations refine rather than reverse this. Seed-null oracles (≈ 4.5 pp) match cross-config plug-in oracles.

Contributions.

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1. Measurement protocol. Seed-null, seed-substitution, SC@kN, margin-split, probe-then-commit tokens, and seed-calibrated P4.
2. C2. Equal-budget CDV: no positive lift; drop-c_2 / quality-weighted ablations show c_2 is a quality confound.
3. C1. Selective plug-in π_{mode} spreads vs. a seed-permutation null where seeds exist.
4. C3. Selector gains vs. SC@N do not survive seed-matched controls on controlled cells; compute cost is $\approx$1.6$\times$ SC@N.

Positioning. We do not claim first discovery of prompt sensitivity (Romanou et al., 2026; Messing, 2026; Polo et al., 2024; Sclar et al., 2024). The delta is modal CoT measurement, a budget-matched pooling test with honest uncertainty, and seed-matched controls that show multi-prompt "selection" gains are easily confounded with denoising. Extended related work: Appendix B.

2. Method

A single sampling corpus supports every quantity offline: no training and no extra generation for CDV.

Notation. For each cell (i, m, c, s) we draw N chains, extract and canonicalise answers, and record k_{imcs} correct. Write $\hat{p}_{imcs} = k/N$ and the Haldane logit $z = \log \frac{k+1/2}{N-k+1/2}$. An elicitation configuration perturbs everything except mathematical content. Core set $\{c_0, c_1, c_2\}$:

- c_0: neutral system ("helpful assistant"), zero-shot CoT, T=0.8;
- c_1: expert-mathematician system prompt; else as c_0;
- c_2: family-minimal / default-system rendering (audited per family).

Parse (c_3), few-shot, and temperature axes are separate; c_5 appears only in temperature-diversified SC (B4).

Modal ceiling. Plurality class $a^\star$ over N samples; $\pi_{\text{mode}}(m, c)$ is the fraction of items with $a^\star$ correct. Gap: pass@N $-$ π_{mode}. We measure π_{mode} ourselves (Brown et al., 2024; Bay & Yearick, 2026).

Supporting GLMM / moments. Crossed binomial GLMM following Messing (2026); Polo et al. (2024); primary variance tables report both method-of-moments shares on Haldane logits and VB-GLMM shares with bootstrap CIs. Seed replicates at c_0 supply the noise floor and r_{mm} for disattenuation $\rho_{\text{disatt}} = \rho_{\text{raw}}/r_{mm}$. Hard-subset Jaccard across configs vs. seeds measures excess instability.

Algorithm 1 Configuration-Diversified Voting (CDV)

Require: item i, model m, token budget B, set $\mathcal{C}_{\text{use}}$
$b \leftarrow B/|\mathcal{C}_{\text{use}}|$
$P \leftarrow \emptyset$
for $c \in \mathcal{C}_{\text{use}}$ do
  draw chains from (i, m, c) until cumulative output tokens reach b; add classes to P
end for
return $\arg\max_a \#\{a \in P\}$

CDV (tested policy). Split completion-token budget B evenly across $\mathcal{C}_{\text{use}} = \{c_0, c_1, c_2\}$, pool answer classes, return plurality (Algorithm 1). Budgets are matched on actual completion tokens (early stop / length accounted for in the offline allocator). Compare to SC and temperature-diversified SC. Primary test P5/F3: plug-in π_{mode}(CDV) vs. $\pi_{\text{mode}}(c_0)$, with item-clustered bootstrap CIs and a pre-specified equivalence margin δ=1 pp (CI-based TOST; reported even when inconclusive).

3. Elementary Observations

We record only elementary facts needed to interpret C1/C2—not a standalone theory contribution. Proofs are immediate; the empirics test the premises.

Definition 1 (Modal hit and ceiling). Fix model m, configuration c, and item i. Let $P_{i,m,c}$ be the distribution over canonical answer classes induced by sampling chains under (m, c). Write mode(P) for a plurality class (ties broken by a fixed rule). The item is a modal hit if mode($P_{i,m,c}$) is correct. Then $\pi_{\text{mode}}(m, c)$ is the fraction of items that are modal hits.

Proposition 1 (i.i.d. resampling cannot move a fixed mode). If chains are drawn i.i.d. from a fixed $P_{i,m,c}$, then as $N \to \infty$ the empirical plurality converges to mode($P_{i,m,c}$). Increasing N cannot change the population mode; it only reduces estimation error of that mode.

Proposition 2 (Mixture modes need not raise modal correctness). Let $\{P_c\}_{c \in \mathcal{C}}$ be configuration-conditional answer distributions and let $Q = \sum_c w_c P_c$ be any mixture with $w_c \geq 0$, $\sum_c w_c = 1$ (equal-budget CDV is the finite-sample analogue). Nothing forces $\mathbf{1}$[mode(Q) correct] $\geq \mathbf{1}$[mode(P_{c_0}) correct] itemwise.

In particular, if moving from c_0 to c promotes an incorrect class into the plurality more often than it demotes one (net-destructive reordering), the mixture can lower modal correctness relative to c_0.

Proposition 3 (Margin gate for a beneficial config change). Fix item i and write $p_c(a)$ for the mass of canonical class a under configuration c. Let $a^\star$ be correct and b any incorrect competitor. A change $c_0 \to c$ can be corrective only if $p_{c_0}(a^\star) \le p_{c_0}(b)$ and $p_c(a^\star) > p_c(b)$ (up to ties). Equivalently, a flip requires the ℓ_∞ change in class masses to exceed half the source top-two margin $m_{c_0} = (p_{c_0}(a_{(1)}) - p_{c_0}(a_{(2)}))$. Corrective mass is therefore concentrated on small-margin items—the regularity we measure descriptively—while destructive flips remain free whenever the correct class is only narrowly modal under c_0. The proposition is symmetric in direction and does not by itself explain CDV's sign.

Remark 1. C1 asks whether $P_{i,m,c}$ itself changes with c enough that $\pi_{\text{mode}}(m, c)$ moves. C2 asks whether a particular mixture policy improves π_{mode}. Proposition 1 explains why C2 is interesting; Propositions 2 and 3 explain why C2 can fail even when C1 holds, and where reordering should concentrate. Our experiments instantiate both. These are interpretive elementary facts, not a standalone theory contribution.

4. Experimental Setup

Models. Qwen/Qwen2.5-{1.5B,3B,7B}-Instruct (Qwen Team, 2024; Yang et al., 2024) and meta-llama/Llama-3.2-3B-Instruct. Inference with Hugging Face float16, device_map=auto, on 1–2× NVIDIA T4 (16 GB). Precision control: NF4 4-bit on 1.5B/3B MATH-500 only.

Benchmarks. MATH-500 (HuggingFace H4, 2024) (headline; 200-item subsample), GSM8K (Cobbe et al., 2021) (secondary; contamination risk (Zhang et al., 2024)), and GSM-Symbolic main/p2 (Mirzadeh et al., 2025) (150 items; stress test). Fixed item-index files are shipped with the harness.

Sampling. Strategy configured_sampling, N=24 chains/item (O2: N=64 on Qwen2.5-3B×MATH-500×$\{c_0..c_3\}$), T=0.8, top_p=0.95, max_new_tokens=1024. Seeds $\{0, 1, 2\}$ at c_0 for Qwen ladder noise floors; headline table uses seed 0. Answers: extract then symbolic canonicalisation; extraction-failure rates logged in T1/T2.

Corpus. 67 run directories (61 fp16 + 6 4-bit); tab1_setup.csv lists 66 fp16 graded rows used in the primary tables. Analysis: python -m src.analysis.paper with VB-GLMM and item-clustered bootstrap (B=1000). Numbered figures/tables plus depth-stability, transition-persistence, selection, and C3-control CSVs ship in paper_artefacts/ and paper/tables/.

Table 1. Result card (plug-in maj@N unless noted).

ID	Comparison	Finding
C1	Config π_{mode}	2/6 cells survive seed-permutation P4
C2	CDV vs. SC	$\Delta \approx -0.7$ pp; CI crosses 0
C3	Cond. div. vs. SC@N	+2.3 pp (14 cells)
C3ctl	Seed-sub / SC@3N	Indistinguishable on 6 cells
Null	Seed oracle	≈4.5 pp ≈ config oracle

Estimand vs. plug-in. The population target is $\pi_{\text{mode}}(m, c)$ under $P_{i,m,c}$. Reported tables are maj@N plug-ins. Depth-stability curves (prefix-N agreement with full-N modes on O2 cells) quantify how often the estimated mode flips as N grows.

Predictions (protocol in code supplement). Confirmatory: P4 (plug-in π_{mode} range >3 pp); P5 (token-matched CDV does not beat SC). Exploratory: whether conditional diversify beats SC@N; whether that lift survives seed-substitution / SC@3N; P1, P7, oracle-recovery %, transfer panels. We do not claim a public registry timestamp.

5. Results

Unless noted, numbers are maj@N plug-ins for π_{mode} at seed 0 from tab2_main_accuracy.csv, with item-clustered bootstrap CIs (B=1000) on primary CDV and selection contrasts. For Qwen2.5-3B×MATH-500 the loader keeps the deeper O2 cells (N=64) over N=24 duplicates.

Reading order. We report C1 (where ceilings move), C2 (equal-budget CDV negative), C3 (selection with seed-substitution / SC@3N controls), then the descriptive reordering panel. Supporting variance/transfer panels are in Appendix A. Scope is the models, prompts, and benchmarks below—not a universal claim about all configuration diversity.

5.1. Modal ceilings across configurations (Q1)

P4 is supported on several cells, not universally. Spreads exceed 3 pp on 1.5B/7B GSM8K (both 6.5 pp) and on GSM-Symbolic main for all three Qwen sizes (3.3 / 10.7 / 13.3 pp), plus 7B Symbolic p2 (6.0 pp). 3B GSM8K (0.5 pp) and Llama GSM8K (2.9 pp) miss the gate; MATH-500 aggregate ranges stay ≤ 3 pp.

Table 2. Plug-in π_{mode} (%; maj@N) by configuration, seed 0, fp16. N=24 except Qwen2.5-3B×MATH-500 (N=64, O2). Range = max − min over c_0–c_2 in percentage points. Bold: P4 supported (> 3 pp).

Model	Dataset	c_0	c_1	c_2	Range
Qwen2.5-1.5B	GSM8K	74.5	73.5	68.0	6.5
Qwen2.5-1.5B	MATH-500	56.0	54.0	57.0	3.0
Qwen2.5-3B	GSM8K	91.0	90.5	91.0	0.5
Qwen2.5-3B	MATH-500†	76.0	76.5	75.5	1.0
Qwen2.5-7B	GSM8K	94.5	93.5	88.0	6.5
Qwen2.5-7B	MATH-500	82.0	80.0	81.5	2.0
Llama-3.2-3B	GSM8K	89.9	88.5	87.0	2.9
Llama-3.2-3B	MATH-500	59.0	59.5	58.5	1.0
Qwen2.5-1.5B	Sym. main	60.7	57.3	59.3	3.3
Qwen2.5-1.5B	Sym. p2	29.3	27.3	27.3	2.0
Qwen2.5-3B	Sym. main	90.0	82.7	79.3	10.7
Qwen2.5-3B	Sym. p2	66.0	65.3	64.0	2.0
Qwen2.5-7B	Sym. main	87.3	81.3	74.0	13.3
Qwen2.5-7B	Sym. p2	81.3	80.7	75.3	6.0

†O2 deep-N cells retained over N=24 duplicates.

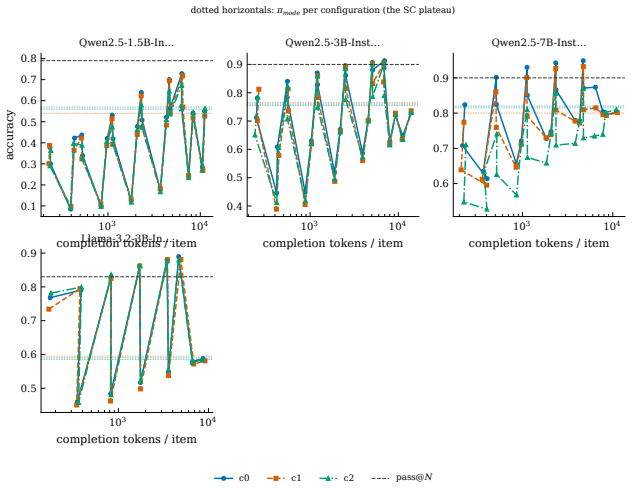


Figure 1. Modal ceiling vs. coverage (artefact F4).

Seed-calibrated P4. On the six multi-seed cells, mean c_0 seed-to-seed π_{mode} range is ≈1.5 pp vs. mean config range ≈3.3 pp. A seed-null permutation (resample three c_0 seed plug-ins; one-sided p= Pr[seed-range ≥ config-range]) leaves only 2/6 cells that both exceed 3 pp and reject the seed null at α=0.05 (1.5B and 7B GSM8K). 3B MATH-500 has seed range > config range. Table 2 is therefore a plug-in comparison; Symbolic's large spreads still lack seeds. Large spreads are often $c_0 > c_1, c_2$ quality gaps (Figure 1).

5.2. Equal-budget CDV does not raise the plateau (C2)

O2 on Qwen2.5-3B×MATH-500 at N=64 gives maj@64 of 76.0 / 76.5 / 75.5 / 76.4% under c_0–c_3. Equal-budget token-matched CDV over $\{c_0, c_1, c_2\}$ (Figure 2; Table T5) shows no positive lift: mean $\Delta \approx -0.7$ pp across 14 cells at the largest matched budget; a pooled normal approximation gives CI $\approx[-1.7, +0.3]$ pp. Per-cell TOST at δ=1 pp is inconclusive everywhere. CDV stays near SC and below oracle best-single / coverage bounds. Some configs (notably c_2) are weaker than c_0 on high-spread cells, so pooling mixes unequal qualities; we do not claim all diversity schemes fail—only that this equal-budget allocator does not demonstrate a lift. Title/conclusion language matches this weaker claim.

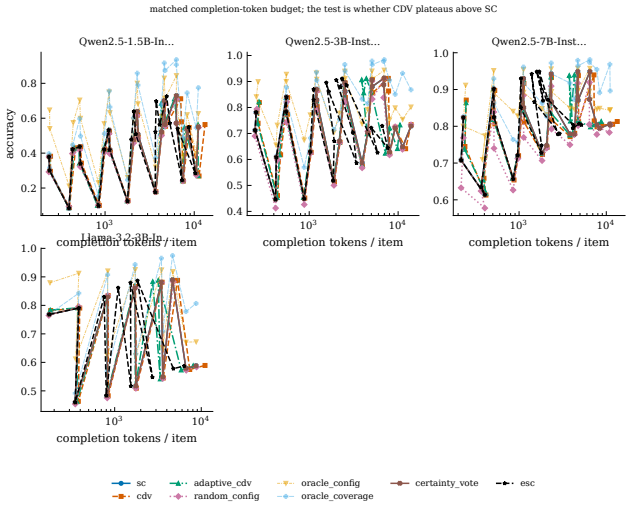


Figure 2. CDV vs. self-consistency under token-matched budgets (F5).

5.3. Held-out selection and seed-matched controls (C3)

Setup. Pre-specified conditional_diversify (threshold fit on train; five overlapping 40% holdouts): use c_0 when its vote margin ≥ τ, else plurality over per-config modes.

Apparent gain (14 cells). Vs. SC@N: mean ≈+2.3 pp (cell-bootstrap CI $[+1.1, +3.7]$). Beats val-selected single-config and uniform mode-vote on average.

Seed-matched controls (6 cells today). c_0 seeds $\{0, 1, 2\}$ exist for Qwen×{GSM8K, MATH-500} only—not Llama or GSM-Symbolic (largest C1 spreads). Replace $\{c_0, c_1, c_2\}$ by c_0 seeds and compare to SC@3N (pooled c_0 seed samples): Both key residuals cross zero: we cannot distinguish configs from seeds at this power. We do not claim proven equivalence. A 16-run seed-expansion queue (Symbolic + Llama, ≈45 GPU-h) is shipped to close the eight uncontrolled cells; analysis scripts auto-expand when seeds land.

Table 3. C3 contrasts (pp). Seed controls: six multi-seed cells; CIs are cell-bootstrap.

Contrast	n	Mean	95% CI
CD−SC (all 14)	14	+2.29	[+1.07, +3.70]
CD−SC (seed 6)	6	+1.58	[+0.42, +2.89]
CD−SC (no-seed 8)	8	+2.82	[+0.94, +4.93]
Seed-sub−SC	6	+2.01	[+0.61, +3.52]
SC@3N−SC	6	+1.20	[+0.22, +2.18]
CD−seed-sub	6	−0.42	[−1.20, +0.28]
CD−SC@3N	6	+0.39	[−0.11, +0.89]

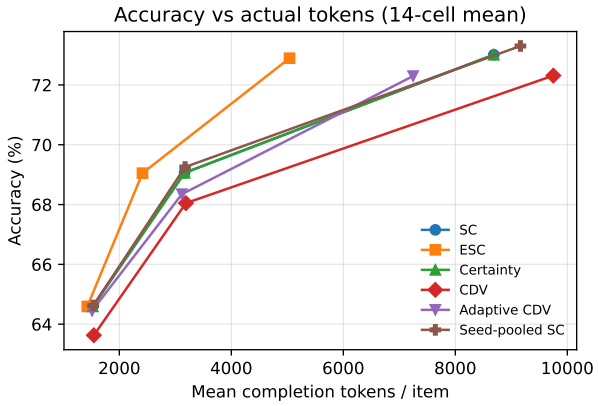


Figure 3. Accuracy vs. mean completion tokens (14-cell mean; offline pool).

Independent margin-split. Estimating the c_0 margin on one half of samples and deciding on the other half still yields CD−SC ≈+3.2 pp (CI [+1.4, +5.5]): the lift is not an artefact of reusing the same draws for margin and vote.

Compute: probe-then-commit tokens. Always sample c_0; add c_1/c_2 only when margin$<\tau$. Mean over 14 cells: early-exit fraction ≈0.74; token cost ≈1.60× SC@N and ≈0.53× always-3. Figure 3 places SC, ESC, certainty-weighted voting, and CDV on an accuracy-vs-actual-tokens curve: ESC matches SC accuracy at substantially fewer tokens; equal-budget CDV does not dominate the frontier.

Quality-matched ablations. Dropping weak c_2 raises CD−SC to ≈+3.3 pp; quality-weighted CD yields only ≈+0.7 pp. Sample-level mixture of the 3-config pool does not beat SC@3N on seed cells (Δ≈−0.3 pp). Thus c_2 is a quality confound for unweighted pooling, not free diversity.

Seed-null oracles. c_0 seeds as fake configs: mean oracle gap ≈4.5 pp, matching cross-config plug-in oracles (Table 4).

Table 4. Seed-null plug-in oracle gap (pp): $\max_{\text{seed}} -\text{seed}\,0$.

Model	Dataset	Gap	n
1.5B	GSM8K	11.1	199
3B	GSM8K	2.5	200
7B	GSM8K	1.0	200
1.5B	MATH-500	6.5	199
3B	MATH-500	3.5	200
7B	MATH-500	2.5	200
Mean		4.5	

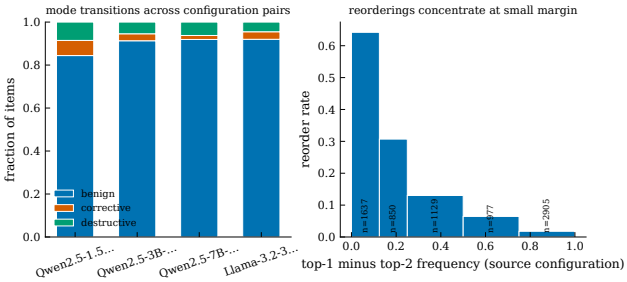


Figure 4. Mode reordering: corrective vs. destructive (F6).

Implication. Withdraw elicitation-headroom claims. Report seed-null, seed-substitution, SC@kN, margin-split, and actual-token cost—on the same cell set as the headline gain—before attributing multi-prompt lifts to elicitation.

5.4. Mode reordering (Q2)

Pairs are unordered combinations evaluated in lexicographic order ($c_A \to c_B$) (so rates are not symmetric by construction). Mean over pairs: reorder 21.6%, corrective 4.0%, destructive 6.4% (Figure 4; Proposition 3). Reorderings concentrate on small source margins. This analysis is descriptive—not a proven causal account of CDV—and partly reflects that stronger configs appear earlier in the lex order. Depth persistence: transition-persistence CSV on O2 cells.

5.5. Variance, transfer, precision, downstream (Q4)

VB-GLMM decomposition (Figure 5; Table T3): item and model dominate; item × config share is 0.82% on GSM8K (CI [0.81%, 1.74%]) and 0.32% on MATH-500 (CI [0.22%, 0.65%]), so P1 is unsupported—ceiling gaps need not imply a large crossed variance share when mass shifts near the decision boundary. Transfer/overlap panels and Symbolic spreads are in Appendix A (F2, F3, F7) and Tables T4–T6. GSM-Symbolic lacks a fully crossed four-model design (no

Llama on that arm), so its variance block is omitted.

Precision (P7; exploratory until ρ_{disatt} lands). On MATH-500, 4-bit drops mean π_{mode} sharply vs. fp16 ($\Delta \approx -16$ pp pooled in T7) while item × config share moves by <0.2 pp. Without 4-bit ρ_{disatt}, P7 remains unsupported (Figure 9); we do not treat share-only agreement as confirmatory.

Downstream (P6). Table T6: where adaptive gains are reportable, transferring difficulty from another configuration often retains <1 of the aware-allocation gain (sometimes negative on 1.5B GSM8K; ≈ 0.94 on 3B GSM8K).

6. Discussion

Joint outcome. Pooling fails to lift SC; a margin-aware rule appears to help vs. SC@N; on the six controlled cells that help is not distinguishable from seed-matched extra samples. That conjunction is the paper—not a positive prompt-selection method. The eight uncontrolled cells caution against over-generalising the denoising attribution.

On reordering and CDV. Mean corrective / destructive rates are 4.0% / 6.4% (Figure 4), but the pair ordering is lexicographic and asymmetric, so these rates are descriptive—not a causal account of C2. Oracle rows in Table T5 are plug-in maxima inflated by the same finite-N noise the seed null exposes; we do not read them as untouched elicitation headroom.

Where C1 is strong vs. weak. Spreads are largest under distribution shift (GSM-Symbolic) and on some GSM8K rungs; MATH-500 aggregate ranges stay ≤ 3 pp. We treat Symbolic as a ceiling stress test and MATH-500 as the place where, for our core prompts, ceilings are comparatively stable. Scale is non-monotone (3B GSM8K 0.5 pp vs. 1.5B/7B 6.5 pp).

Small variance shares are compatible with C1. GLMM item × config shares <1% (point) do not contradict Table 2: reordering near the decision boundary can move maj@N without inducing a large crossed random effect of the kind BrittleBench/TEE emphasise on deterministic suites. Variance decomposition remains supporting measurement, not the headline.

What we are not claiming. We do not claim that prompt selection recovers elicitation headroom, that CDV is equivalent to SC, or frontier generality. We claim a careful negative on equal-budget pooling and a measurement result: without seed/depth controls, multi-prompt "selection" gains are easy to misattribute.

7. Limitations

- Scale. Models fit on ≤ 2×16 GB GPUs; frontier / reasoning-specialised behaviour may differ.
- Configuration sample. Three core prompts are not a census; C2 is for equal-budget CDV on $\{c_0, c_1, c_2\}$; C3 policies are exactly the held-out selectors we implement (not an exhaustive search over all rules).
- Semantics preservation is argued via axes that avoid rewriting the problem; residual role / length / format shift is possible. We log per-config extraction failures (T1/T2).
- Finite N / plug-in modes. N=24 (mostly); O2 uses N=64 on selected MATH-500 3B cells. Depth-stability and transition-persistence tables quantify mode-label sensitivity to sampling depth.
- Uncertainty / power. Primary contrasts use item-clustered bootstrap CIs (B=1000) and VB-GLMM shares. CDV−SC TOST at δ=1 pp is inconclusive: we establish no positive lift, not equivalence. Some VB fits emit non-convergence warnings; we report shares alongside moment estimates in T3.
- Benchmarks. MATH-500/Symbolic lead; GSM8K is secondary (contamination). Symbolic lacks a fully crossed Llama arm.
- Canonicalisation. Symbolic checker errors enter π_{mode}; we log extraction failures.
- P7. 4-bit ρ_{disatt} required before calling P7 supported (exploratory until then).
- C3 controls. Seed-substitution / SC@3N cover six Qwen GSM8K/MATH-500 cells (three seeds). Config−seed CI $[-1.20, +0.28]$ pp and config−SC@3N CI $[-0.11, +0.89]$ pp both cross zero. Llama and GSM-Symbolic seeds are queued (configs/seed_expansion/; ≈45 GPU-h for P0) but not yet graded. Probe-then-commit token ratios are offline estimates.
- Seed null / P4. Oracle gaps and seed-permutation P4 use three c_0 seeds; deeper seeds ($\{3, 4\}$) are queued as P1.

8. Conclusion

Equal-budget prompt pooling does not beat self-consistency under our CIs. A margin-aware aggre-

gator can beat SC@N, but on controlled cells we cannot distinguish that lift from seed-substitution / SC@3N; probe-then-commit still costs ≈1.6× SC@N tokens. Plug-in oracles and uncalibrated P4 spreads are inflated by finite-N noise. The practical message is methodological: do not ensemble prompts by default, and do not credit multi-prompt selectors without seed-null, seed-substitution, SC@kN, margin-split, and actual-token controls on the same cell set as the headline gain.

Impact Statement

This work studies evaluation and test-time aggregation for open-weight math reasoning models. Potential impacts are improved honesty in reporting reasoning benchmarks and more careful use of difficulty labels in curricula. We do not release new high-risk capabilities; experiments are inference-only on public benchmarks. Misuse risk is low relative to generative systems; the main risk is over-interpreting C2 as "prompts never matter" (contradicted by C1) or over-interpreting C3 as elicitation recovery (contradicted by seed-substitution and SC@3N).

Reproducibility Statement

All experiments are inference-only on open-weight models. Harness, configs, and analysis code accompany the submission; artefacts reproduce via scripts/split_precision.py then python -m src.analysis.paper. Exact run inventory: tab1_setup.csv. Figure/table files: paper/figures/, paper/tables/.

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A. Additional Figures

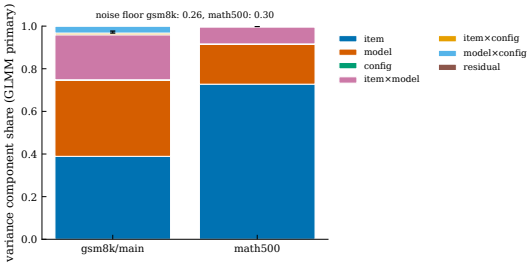


Figure 5. Variance components (F1; VB-GLMM shares with moment comparison in T3).

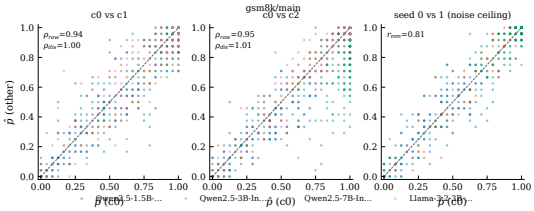


Figure 6. Difficulty transfer across configurations (F2).

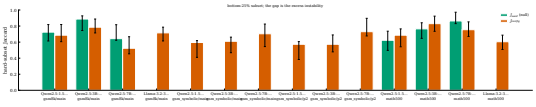


Figure 7. Hard-subset overlap across configs vs. seeds (F3).

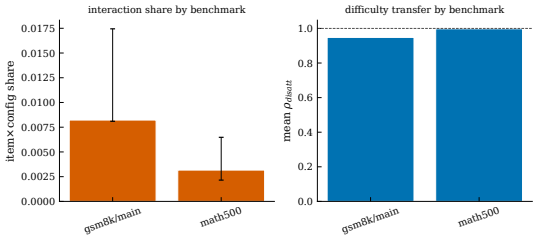


Figure 8. GSM-Symbolic shift / ceiling spreads (F7).

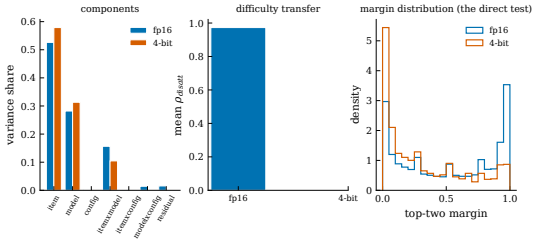


Figure 9. fp16 vs. 4-bit control (F8).

B. Extended Related Work

B.1. Chain-of-thought and its structural variants

Chain-of-thought prompting (Wei et al., 2022) and its zero-shot form (Kojima et al., 2022) established that eliciting intermediate steps improves reasoning accuracy, and self-consistency (Wang et al., 2023) established that sampling several chains and taking a plurality vote improves it further. A family of structural elaborations followed: least-to-most decomposition (Zhou et al., 2023), tree- (Yao et al., 2023) and graph-structured (Besta et al., 2024) search over partial thoughts, and program-mediated reasoning that offloads arithmetic to an interpreter (Chen et al., 2023a; Gao et al., 2023). Later work tempered the picture: Sprague et al. (2025) find that CoT's benefit is concentrated on mathematical and symbolic tasks and is small or negative elsewhere, and much of the apparent advantage of elaborate search structures narrows once sampling budget is matched. What remains unresolved: the community still lacks a principled account of which items benefit from which structure, and almost all such accounts are stated in terms of a per-item difficulty that can shift under elicitation and under finite-N mode estimation noise.

B.2. Test-time compute scaling and its ceilings

Our benchmarks are MATH-500 (HuggingFace H4, 2024) (headline) and GSM-Symbolic (Mirzadeh et al., 2025), with GSM8K (Cobbe et al., 2021) as a secondary replicate. Verifier-reranked sampling (Cobbe et al., 2021), process supervision (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024), and compute-optimal allocation between sequential revision and parallel sampling (Snell et al., 2025; Wu et al., 2024a) form the backbone of test-time scaling; Brown et al. (2024) showed that coverage—the probability that at least one of n samples is correct—keeps rising over orders of magnitude of n, while plurality voting and learned reward models plateau. Muennighoff et al. (2025) showed that a very simple budget-forcing scheme captures much of the benefit. Surveys now organise the area along what/how/where/how-well axes (Zhang et al., 2025) and by subproblem topology (Yang et al., 2025), and a large matched-condition study finds that no single strategy dominates across models and difficulty regimes (Agarwal et al., 2025).

The most directly relevant recent development is the explanation of the plateau. Brown et al. (2024) showed coverage keeps rising while plurality voting plateaus; subsequent work formalises a modal ceiling: plurality voting converges to the mode, so selection accuracy saturates at π_{mode}; the gap to rising coverage is the identifiability gap (Feng et al., 2025; Cordero-Encinar & Duncan, 2025; Huang et al., 2025b). What remains unresolved: all of this conditions on a per-item answer distribution under a fixed elicitation setting. Bay & Yearick (2026) estimate run-to-run variance within a

configuration ($\rho_w \approx 0$) but never vary the prompt; whether π_{mode} is configuration-relative is the step we test.

B.3. Prompt sensitivity and variance decomposition

Three 2026 lines establish that per-item outcomes are prompt-attributable and that evaluation pipelines must model crossed facets. BrittleBench (Romanou et al., 2026) decomposes binary correctness variance into task difficulty and semantics-preserving prompt sensitivity on an item × perturbation matrix; inference is deterministic (log-prob scoring), so the inference-variance term vanishes. TEE (Messing, 2026) fits crossed random effects—including item × prompt—by REML and uses decision studies on aggregate benchmark scores. PromptEval (Polo et al., 2024) estimates performance distributions across prompt templates with an IRT-style model on the joint (item, template) matrix. Earlier aggregate work (Sclar et al., 2024; Mizrahi et al., 2024; Alzahrani et al., 2024) showed format and template choice move rankings by large margins.

What remains unresolved is the intersection with sampled chain-of-thought on free-generation math and with the mode. BrittleBench does not report π_{mode}, reorder rates, or CDV; TEE's estimand is $\text{Var}(\hat{\theta})$ at benchmark scale—item × prompt is a nuisance that averages away in aggregate CIs, whereas adaptive test-time compute, hard-subset curation, and IRT consumers use per-item parameters one at a time. Żatuchin (2026); Zhu et al. (2026) show that under G-theory facet decompositions, repeats are the cheap facet and paraphrases/models buy more reliability per query—the allocation-theoretic sibling of "i.i.d. resampling cannot move the plateau," but on brand sentiment without modes or reasoning. Zhou et al. (2026b) establish unstable hardness labels along the stochastic-vs-deterministic axis; we study the configuration axis and whether it reaches the selection bottleneck. Adjacent aggregation devices—paraphrase voting (Chen et al., 2024a; Zhou et al., 2024)—are baselines, not the measurement target.

B.4. Long chain-of-thought, RL for reasoning, and the elicitation debate

Reasoning-specialised models trained with reinforcement learning on verifiable rewards (DeepSeek-AI, 2025; Shao et al., 2024) produce long chains-of-thought and large gains, and distillation transfers much of the behaviour into small models with remarkably little data (Muennighoff et al., 2025; Ye et al., 2025). A pointed debate followed about what RL actually contributes. Yue et al. (2025) argue via pass@k analysis that RLVR largely sharpens the base model's existing distribution rather than extending its reach, and Dang et al. (2025) formalise a related bias–variance decomposition of pass@k, showing that supervised fine-tuning improves pass@1 while collapsing generation diversity and degrading pass@k, and that weight interpolation partially recovers both. Theoretically, self-improvement has been framed as sharpening a fixed base policy (Huang et al., 2025a), and inference-time methods have been recast as sharpening the answer marginal directly (Arzhantsev et al., 2026). What remains unresolved: the sharpening framing implies that the ordering of candidate answers is doing all the work, but the stability of that ordering—under scale, under post-training, and under elicitation—has not been measured. Our mode-reordering analysis (Section 5.4) is the missing measurement for the elicitation axis.

B.5. Faithfulness, premature commitment, and what the chain is for

A parallel literature asks whether the chain reflects the computation. Turpin et al. (2023) show that models silently follow biasing features in the context while producing unrelated rationales; Lanham et al. (2023) introduce the truncation and corruption probes that have become standard; and Lyu et al. (2023) propose constrained formulations that are faithful by construction. Latent-space variants ask whether the tokens are needed at all (Hao et al., 2024; Pfau et al., 2024; Goyal et al., 2024). Recent work sharpens the commitment question: Gai et al. (2026) probe at evenly spaced truncation points, define a premature-confidence score from the resulting confidence trajectory, show it predicts logical flaws, and mitigate it with a shaped RL objective; Cox et al. (2026) show that a linear probe on pre-CoT activations predicts the final answer with high AUC and that the direction is causal under steering. Jiang et al. (2025) show the converse decoupling—answers that stay correct while the rationale collapses under imperceptible input perturbation. What remains unresolved: these results establish that a model often has an answer in mind early, but they do not connect that to the selection bottleneck at the heart of test-time scaling. Early commitment is one candidate mechanism for genuine mode shifts; our seed-null and seed-substitution results show that many apparent shifts are also consistent with finite-N estimation noise, so the mechanism remains open.

B.6. Efficiency, overthinking, and adaptive compute

Reasoning models overspend on easy items (Chen et al., 2024c) and under-explore on hard ones (Wang

et al., 2025a); surveys catalogue the resulting efficiency literature (Sui et al., 2025). Length-controlled policy optimisation (Aggarwal & Welleck, 2025) and budget-forcing (Muennighoff et al., 2025) give explicit control over chain length. On the parallel axis, adaptive-consistency (Aggarwal et al., 2023) and early-stopping self-consistency (Li et al., 2024) halt when the vote is decided; Certaindex/Dynasor (Fu et al., 2024) generalise this to an algorithm-agnostic "answer stabilisation" signal used for scheduling; Blend-ASC (Feng et al., 2025) allocates samples across questions to optimise sample efficiency; Cordero-Encinar & Duncan (2025) provide sequential certificates; and Bilal et al. (2026) study difficulty-conditioned budget allocation directly. What remains unresolved: every one of these estimates a per-item quantity in one setting and consumes it in another, and none validates that the estimate transfers. This is the applied consequence we measure.

B.7. Robustness, perturbation, and the reasoning-versus-memorisation debate

Mirzadeh et al. (2025) introduced templated variants of GSM8K and reported both high variance across instantiations of the same template and a large drop when an irrelevant clause is added (GSM-NoOp), arguing for pattern-matching over formal reasoning. Contamination audits (Zhang et al., 2024), counterfactual evaluation (Wu et al., 2024b), premise-order effects (Chen et al., 2024b) and the reversal curse (Berglund et al., 2024) point the same direction, while compositionality analyses (Dziri et al., 2023) give a mechanistic account of multi-step failure. The debate is live: puzzle-complexity critiques and their rebuttals continue, and 2026 replications of GSM-NoOp on frontier models suggest that a substantial part of the original effect reflects models reasonably treating ambiguous clauses as informative rather than failing to reason. Follow-up work finds that program-mediated reasoning does not repair GSM-Symbolic robustness (Kutakh, 2026). What remains unresolved: these studies compare accuracy across perturbations of the problem. Ours holds the problem fixed and perturbs only the elicitation, isolating a source of instability that cannot be blamed on the item—subject to the seed-matched controls in the main text.

B.8. Self-correction, verification, and aggregation beyond frequency

Iterative refinement with self-generated feedback (Madaan et al., 2023) and multi-agent debate (Du et al., 2024) were early attempts to convert extra inference into extra accuracy without a verifier, but intrinsic self-correction without external feedback does not reliably improve reasoning (Huang et al., 2024; Kamoi et al., 2024), which is part of why the field turned to verifiers and to better aggregation. Beyond plain plurality voting there is now a dense cluster: universal self-consistency (Chen et al., 2023b), program-verified voting (Toh et al., 2024), semantic weighting of reasoning paths (Knappe et al., 2024), confidence-informed voting and its efficient variant (Petullo et al., 2026), learned rerankers over answer-level features (Marina et al., 2026), radial consensus scoring for best-of-N (Nguyen et al., 2026), and best-of-majority hybrids (Di et al., 2025). Diversity-side interventions are covered in Appendix B.3. What remains unresolved: this literature searches for a better selector given a fixed sample pool. We ask a prior question—what determines the pool's mode in the first place—and show that the answer includes a factor the selector literature treats as fixed. The strongest of these methods are our baselines.

B.9. Uncertainty, calibration, and evaluation methodology

Confidence signals for reasoning include self-evaluated correctness (Kadavath et al., 2022), semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024), verbalised confidence (Lin et al., 2022; Tian et al., 2023), and vote share itself. On the evaluation side, the field has become considerably more careful: Miller (2024) argues for routine interval reporting, Hochlehnert et al. (2025) document how much of recent reported progress on small reasoning benchmarks is seed variance, Hariri et al. (2026) propose Bayesian posterior reporting in place of pass@k, Dragoi et al. (2025) show that pass@k is biased toward low-reliability "lucky hits" and propose a reliability-controlled alternative, and Wang et al. (2025b) show that final-answer matching admits false positives at a rate that materially lowers the true inference-scaling ceiling. Psychometric approaches fit item difficulty explicitly and use it for adaptive testing and routing (Li et al., 2025; Chen et al., 2025; Song et al., 2025; Zhou et al., 2026a), and Kim et al. (2025) show that errors are strongly correlated across models, capping ensembles—an effect formalised as a co-failure ceiling (Chen, 2026). What remains unresolved: the psychometric line assumes a unidimensional item difficulty and validates it largely on non-reasoning benchmarks. We test that assumption where it is most consequential, and we adopt the evaluation hygiene this literature demands throughout.